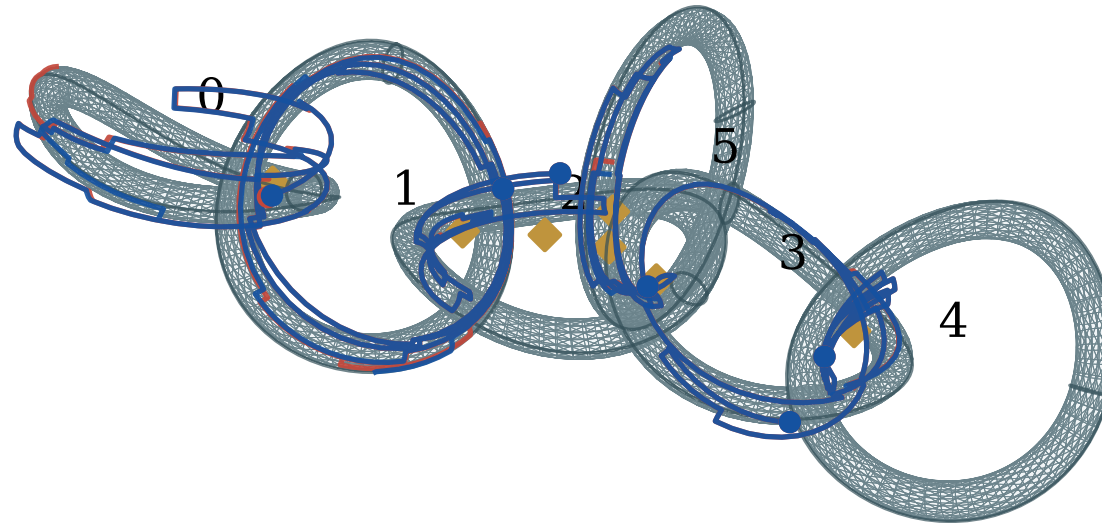


One Holon: linked geometry and circulating fields

Heads and local charts are operations within the same evolving object; its visible geometry is one receiver of that object.



$$\begin{aligned} |\tilde{H}\rangle &\in \Gamma(K), \quad |\tilde{H}\rangle_\alpha = r_\alpha |\tilde{H}\rangle, \quad z = (q, p) \in \mathbb{T}^{12} \\ |\tilde{H}'\rangle &= \Phi_\Theta(|\tilde{H}\rangle), \quad |\tilde{H}\rangle_{n+1} = \Phi_\Theta(|\tilde{H}\rangle_n) \end{aligned}$$

Cycles, joins and field currents belong together.

Six toroidal domains form a branched chain. Neighboring cores have a linking number of magnitude one, verified by their spanning-disk crossings. Their thicker field domains overlap at the gold contact witnesses. The linked carrier is advected by an explicit incompressible twist. Red and blue curves follow two nearby phase states through that moving material; the shown carrier is its final-time cut.

Definition for the supplied analytic phase configuration; established-bounded · computational-witness for the depicted trajectories and geometry. This refines the mathematical design; it is not a reproduced trained-model basin experiment. HNN_FORMULA.md retains the general current/material/incidence law.

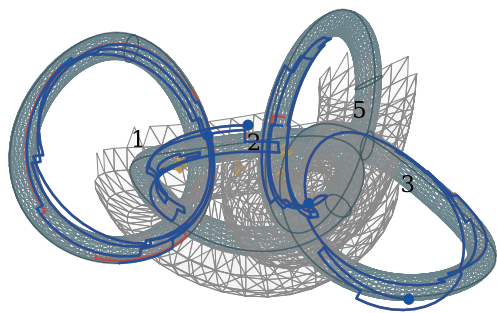
The complete state exceeds the picture.

Each torus has two circulating phase coordinates. The combined reference has twelve phase axes; the 3D embedding displays its constituent cycles and actual field contacts. It is not a twelve-dimensional object flattened into twelve physical planes. Amplitude, constitutive storage and geometry may also vary in the general field specification.

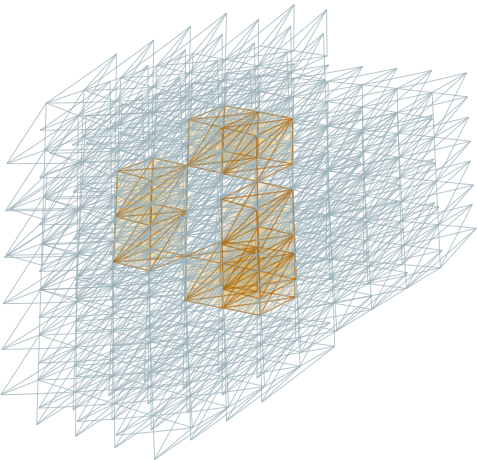
Intersecting fields have shared cells and interior directions

A head is a current/comparison chart of this body. The chart boundary is not a wall between independent models.

Overlapping field domains around a real join



Magnified shared volume · conforming tetrahedra



$$\partial^2 = 0, \quad r_{\alpha\beta} \Big|_{\tilde{\sim}} \Big\langle H \Big\rangle_{\alpha} = g_{\alpha\beta} r_{\beta\alpha} \Big|_{\tilde{\sim}} \Big\langle H \Big\rangle_{\beta}$$

$$F_t(x,y,z) = (x,y\cos(\chi xt) - z\sin(\chi xt), y\sin(\chi xt) + z\cos(\chi xt)), \quad \det DF_t = 1, \quad \chi = \frac{1}{8}$$

An interior, not only an exterior skin.

The cutaway shows the field-support domains around a branch. The magnified view transports one quarter-unit volume complex through that fluid map: 31,440 conforming tetrahedra, including 240 with multiple domain owners. Gold cells carry shared support. Curved edges are advected source edges; common vertices and faces join the complex; the higher cells retain interior variation and their oriented boundary.

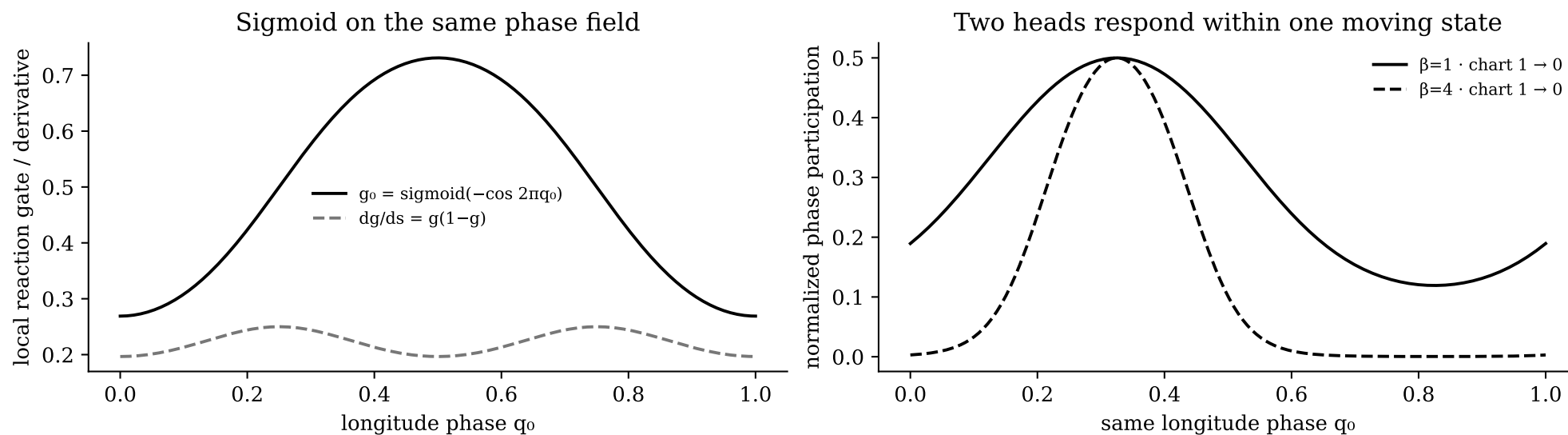
Gluing and physical transport have different roles.

Chart transitions make restrictions compatible. A connection transports current along actual paths and can return nontrivial loop holonomy. Interlinked cores do not alone create contact; the overlap witnesses and the specified constitutive interaction do. The 3D display is an embedded finite realization of these relationships, with its sampling and containment bound retained.

Established-bounded · computational-witness: `intrinsic_holonic_flow/mesh.json`. The inscribed cubes lie within their recorded torus owners by a distance bound; they are subdivided with common incidence. Source owners: `simplicial`, `GradedCausalComplex`, `analytic_field` and `connection/holonomy`.

Softmax and sigmoid respond inside the phase field

Both heads compare currents on the same admitted contacts; the local sigmoid supplies a smooth reaction coefficient.



$$s_{ij}^h = \beta_h \cos(2\pi(q_i - q_j - \varphi_{ij})), \quad a_{ij}^h = \frac{\exp(s_{ij}^h)}{\sum_{k \in E_i} \exp(s_{ik}^h)}$$

$$g_i = \sigma(-\cos(2\pi q_i)), \quad \sigma'(s) = \sigma(s)(1 - \sigma(s))$$

$$\delta a_j = a_j \left(\delta s_j - \sum_k a_k \delta s_k \right), \quad J = \text{diag}(a) - aa^T$$

A head is a field-sensitive operation.

The plotted phase is q_0 in the same twelve-axis state. The two heads use $\beta=1$ and $\beta=4$ on the same contact incidence and oriented phase connection. Their different responses affect the same current; they do not occupy separate parallel bodies. The sigmoid is evaluated on a local phase potential, with its derivative retained.

Normalization retains curvature and participation.

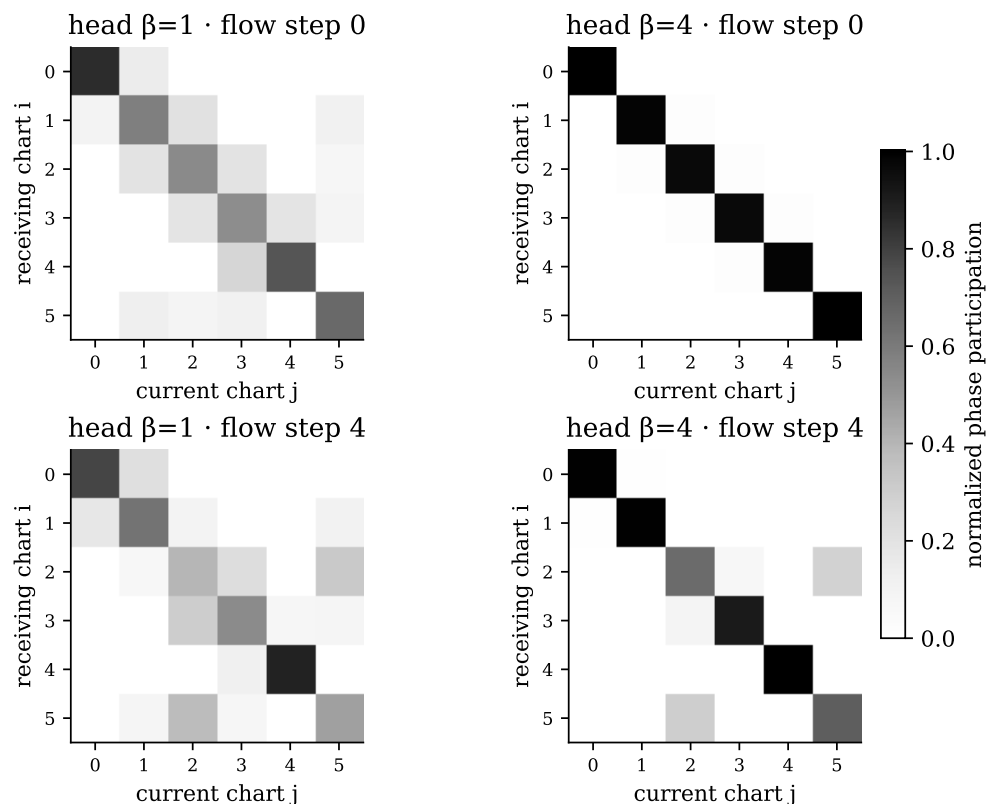
Its differential is $J = \text{diag}(a) - aa^T$, a covariance/Laplacian with constant-shift null direction. The complete transport differential also includes changed values and frame transport. A current can curve, exchange phase and change participation while remaining a section of the same field. Adding contact requires the incidence law as well.

$$\delta Y = \sum_j a_j U_j \delta V_j + \sum_j a_j \delta U_j V_j + \sum_j a_j (\delta s_j - E_a \delta s) U_j V_j$$

Proved-derived · formal-checked: normalized exponential and sigmoid directional derivatives in `HolonicAdjointNormalization`. The plots evaluate the analytic reference numerically; they show its phase-dependent coefficients, not separate head topologies.

State and material variation share one differential

Changing a current changes participation. Learning changes the operator material through the generated result's derivative.



The same two heads at two flow times.

These matrices are numerical receiver faces of the linked object. Their entries change because q changes under the nonlinear operation. The map's coefficients remain fixed in this comparison. Treating this state effect as a learned material update would conflate two derivatives.

$$\delta x' = D_x \Phi_{\Theta} \delta x + D_{\Theta} \Phi_{\Theta} \delta \Theta$$

$$D_{\Theta}(\Phi_{\Theta} \circ \Phi_{\Theta}) = D_x \Phi_{\Theta} D_{\Theta} \Phi_{\Theta} + D_{\Theta} \Phi_{\Theta}$$

In the second equation the outer derivatives are evaluated at the intermediate state. Learning pulls a supplied output covector through this whole composition and uses a declared material metric/update law. The same parameter appearing in several local operations contributes through every use.

The normalized current has an explicit pullback.

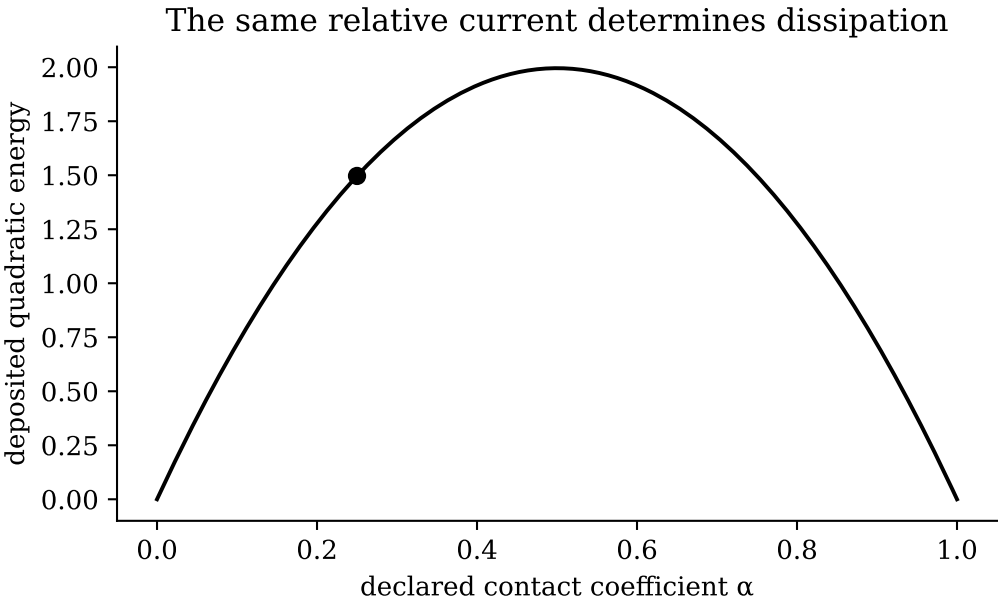
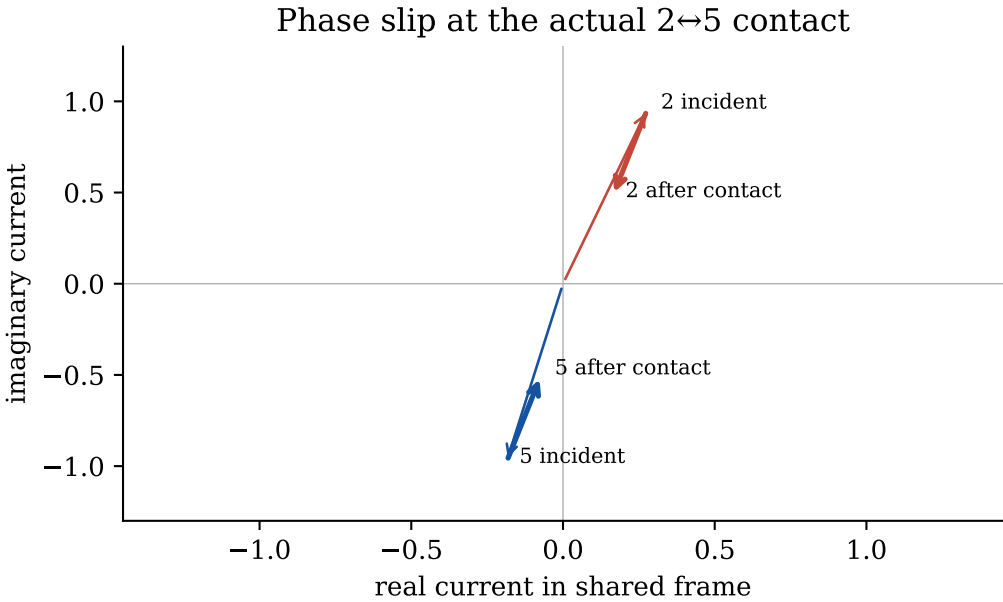
$$Y = aV, \quad G = \frac{\partial \ell}{\partial Y}$$

$$\frac{\partial \ell}{\partial s_{ij}} = a_{ij} \langle G_i, V_j - Y_i \rangle, \quad \frac{\partial \ell}{\partial V} = a^T G$$

Definition for the complete operator variation; proved-derived for the chain rule. The existing Rust NormalizedKernel checks the full tangent/pullback pairing and an observed material update in its separate exact logarithmic diagnostic. That bounded control does not replace the nonlinear model equation or its geometric carrier.

Relative phase determines a contact's dissipative return

The currents below come from the same toroidal branch, expressed in the common frame of its 2↔5 contact.



$$\delta = y - ux, \quad |u| = 1, \quad x' = x + \alpha \bar{u} \delta, \quad y' = y - \alpha \delta$$
$$|x|^2 + |y|^2 - |x'|^2 - |y'|^2 = 2\alpha(1 - \alpha)|\delta|^2$$

Friction responds to a situated difference.

The connection transports x into y 's frame before comparison. At $\alpha=1/4$ their relative slip contracts and the complete quadratic energy difference is deposited. Neither the individual intensity nor a scalar attention coefficient determines this phase-sensitive exchange on its own. The plotted coefficient family uses the same incident currents.

A physical field keeps the receiving modes.

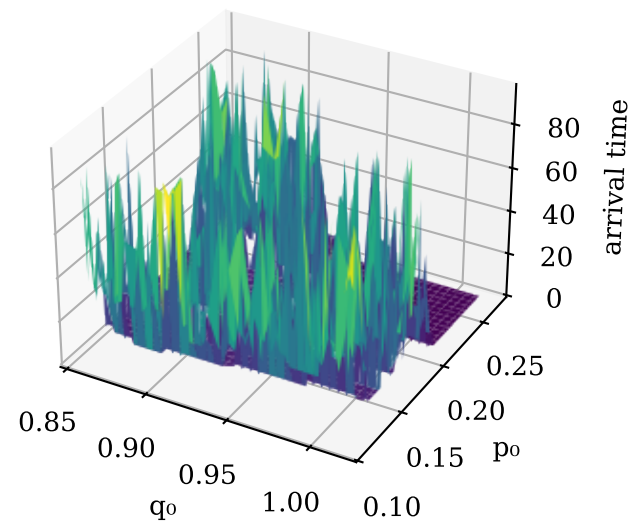
Dissipation changes radial amplitude and deposits energy into internal/thermal modes. Those modes augment the conservative unit-phase chart used for the basin calculation; they are not erased by its phase-only view. HNN's constitutive field and reflected interior already supply the corresponding current, material and storage roles.

Proved-derived for the displayed balance; established-bounded · computational-witness for the local contact evaluated on the selected phase state. The conservative arrival map and this dissipative contact are explicitly different constitutive restrictions of the general field, not an assertion that phase motion alone models all friction.

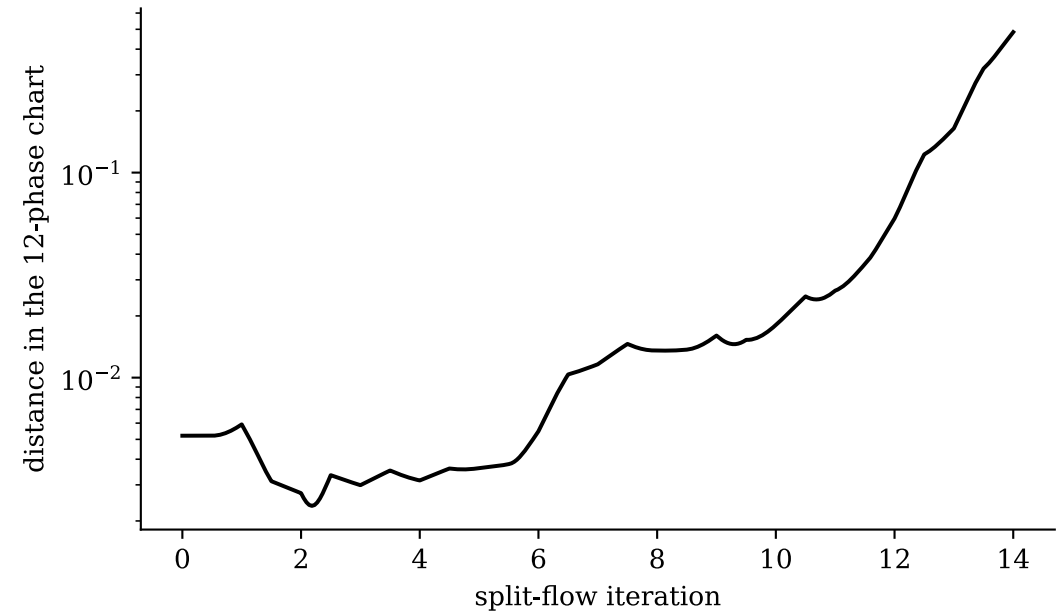
Generation follows curved routes through the whole state

Nearby initial sections can have different arrival times while every head acts within the same continuing field.

A receiver-height surface of the same section



Neighboring starts · arrival 9 versus 96



$$q' = q + \kappa \sin(2\pi p), \quad p' = p - \nabla V(q') \pmod{1}$$
$$\tau_A(z) = \min\{n : \Phi^n(z) \in A\}, \quad A = \{q_0 \in [0.20, 0.34), p_0 \in [0.08, 0.20)\}$$

The complete generator is nonlinear.

V combines the two normalized phase-comparison potentials and sigmoid reaction. Each split subflow is Hamiltonian; their composition preserves phase volume without requiring global contraction. The paired states shown on the linked tori first reach this detector after 9 and 96 iterations. Their separation uses all twelve phase coordinates.

Established-bounded · computational-witness for the finite numerical trajectories. The detector, sample section and 96-step observation window are declared. Gray cells have not arrived within that window. They are retained outcomes, not silently classified as impossible or divergent.

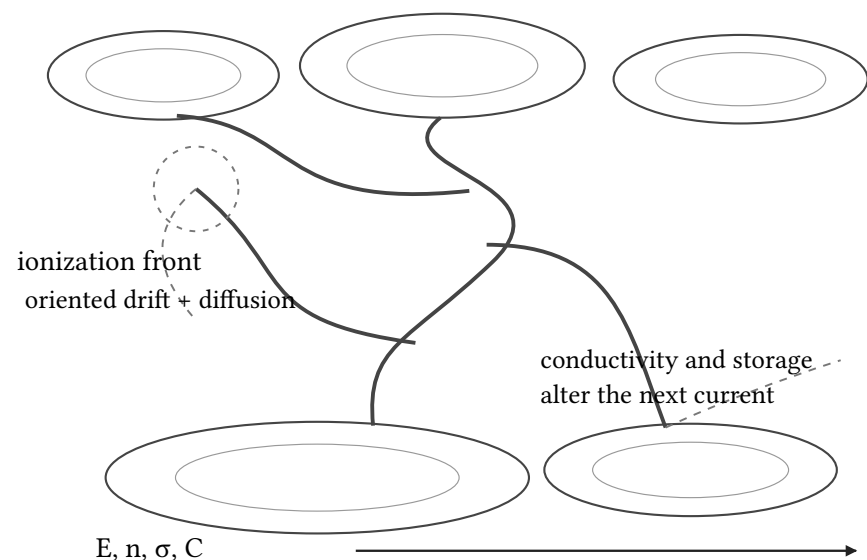
An output does not require the whole body to settle.

A receiver reads the generated activity that reaches it. The plotted height is first arrival, not a physical surface height or proof of a fixed point. Image, acoustic and text decoders read their own structured fields and boundaries; the refinement clock is not universally an output-token position. The full state continues beyond an observation when its application requires it.

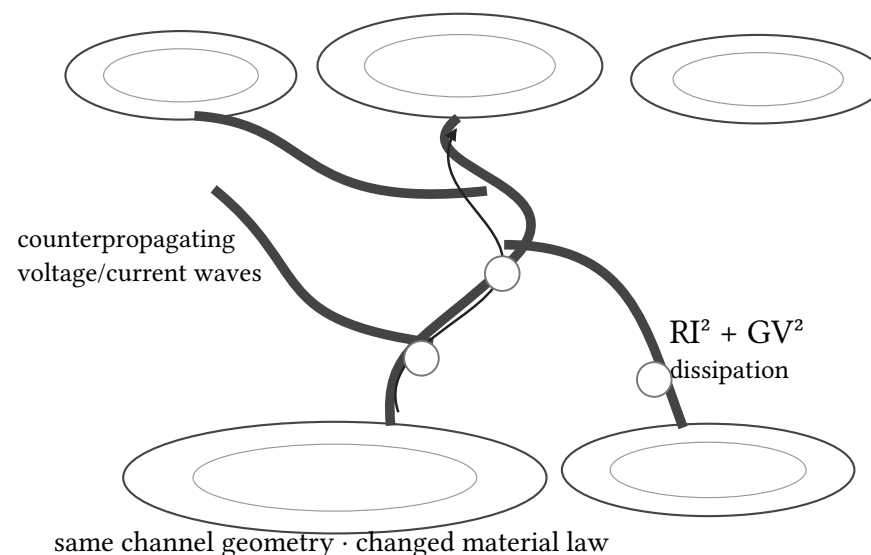
Leader formation changes the available geometric pathways

Material, incidence, current and dissipation are coupled parts of the same field construction.

Leader: material and admissible pathways change



Return: current in the changed material



$$\Gamma_e = n_e u - \mu_e n_e E - D_e \nabla n_e, \quad \partial_t n_e + \text{div} \Gamma_e = S_{\text{ion}} + S_{\text{photo}} - S_{\text{attach}} - S_{\text{recomb}}$$

$$\text{div}(\epsilon E) = \rho, \quad \partial_s V = -\partial_t(LI) - RI + e, \quad \partial_s I = -\partial_t(CV) - GV + i$$

The changing medium determines continuation.

Electron transport, ionization, electric field and boundary conditions determine leader growth. It changes conductivity, storage and channel incidence; the return then propagates in that changed material. The schematic connects the existing conducting-fluid and channel laws. Its branches are annotated relationships, not measured plasma trajectories.

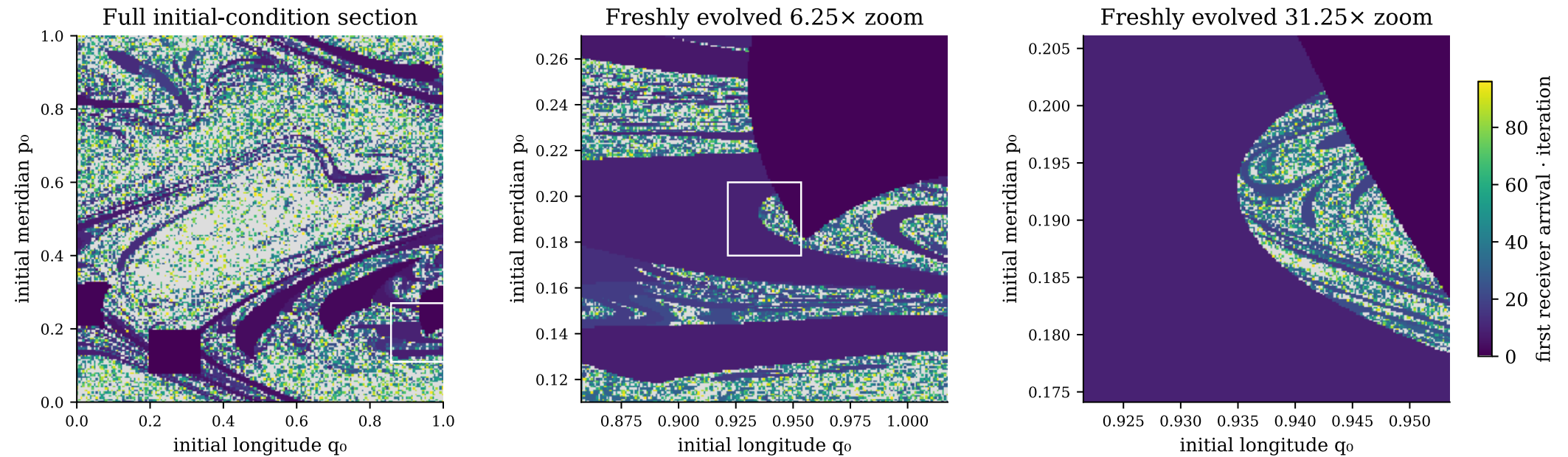
Definition for the physical equations; interpretation for the explicit HNN map between oriented variation, changing incidence and current. Source: FLUID_REFLECTION_AND_CONCENTRATED_INTERIORS.md. This is an extension of the same object, not a topology assigned by a head or layer name.

A fixed-contact phase configuration is one restriction.

The preceding phase calculation holds an already constituted contact complex. The full HNN operation can also change the complex and material through the existing source-conditioned current/variation owners. A local derivative on a fixed support does not perform that transition. The retained interior and higher-cell incidence carry the terms needed to compose it.

Fractal geometry belongs to the nonlinear operation

These are freshly evolved sections and zooms of one twelve-phase map, not copies placed at smaller spatial scales.



$$A_0 = A, \quad A_{n+1} = \Phi^{-1}(A_n) \setminus A, \quad z \in A_n \Leftrightarrow \Phi^n(z) \in A \wedge \forall k < n, \Phi^k(z) \notin A$$

Repeated preimages expose the fine geometry.

Color records first arrival at the same detector. Stretching, folding and repeated restriction produce the interleaved route populations. Each 192x192 panel starts new trajectories on the indicated smaller initial-condition section; it is not an enlarged bitmap or a grid of duplicated models. The other ten initial phase coordinates are held fixed.

The number of material units need not grow.

A fixed-dimensional nonlinear state can have fractal invariant, survivor or arrival-boundary families. Their geometry belongs to the generating operation and its compatible conditions. The old “while the population grows” convention was too narrow. The finite panels here expose multiscale return structure; an infinite-scale dimension theorem requires its own hypotheses and proof.

Proved-derived · formal-checked: HolonicRecurrentEcology.FirstArrival proves the first-hit characterization and chart covariance. Established-bounded · computational-witness for the sampled phase map. The supplied paper’s panels similarly read latent initial-condition sections, but use trained models and decoded settling; this reference uses its own explicit analytic law and first-arrival receiver.

Scale, phase and decoding are operations on the field

A smooth local chart, a fractal family and an emitted image can describe different aspects of the same generative object.

Circulation and analytic phase

$$\psi_i = \exp(2\pi i q_i), \quad U_\gamma = \prod_{e \in \gamma} U_e$$

The two torus cycles, their relative phase and path-ordered transport remain in the source behind an exterior picture. The constitutive currents can excite many modes. A monochrome or intensity receiver discards some of their phase distinctions; it does not define the entire Holon.

Recursive restriction and rebase

$$r_{\text{out}} \circ \Phi_{\text{fine}} = \Phi_{\text{coarse}} \circ r_{\text{in}}$$

A reusable coarse law must commute with the admitted action, or retain its actual defect and relevant unresolved modes. Branching preimages and recursive restrictions need not be a disjoint IFS. Fractal dimension, tensor rank, spatial dimension and the number of displayed axes answer different questions.

Geometric and modal decoding

$$A(x) = \sum_i \psi_i \varphi_i(x), \quad y_R = \rho_R(A)$$

A source-resolved reconstruction produces a wave, image or geometric field. Its interior, surface, phase and intensity views have distinct receiver maps. A single outer isosurface can conceal linked cycles and shared interiors; use the view that exposes the interaction under discussion.

$$d_{\text{box}} = \lim_{\varepsilon \rightarrow 0} \frac{\log N(\varepsilon)}{\log(\frac{1}{\varepsilon})}, \quad f(\varepsilon) \asymp \varepsilon^\alpha, \quad d_{\text{section boundary}} = 2 - \alpha$$

The boundary relation on the right requires its sampling and scaling hypotheses. In this construction the torus surface has dimension 2, the shared volume cells dimension 3, the phase chart dimension 12, and the plotted initial-condition section dimension 2. None of those integers is an estimate of a limiting fractal dimension. The receipt retains finite early/late disagreement across five separations without promoting a finite fit to that limit.

$$x_t = \sqrt{\overline{\alpha}_t} x_0 + \sqrt{1 - \overline{\alpha}_t} \varepsilon$$

$$dx = [f(x, t) - g(t)^2 \nabla_x \log p_t(x)] dt + g(t) d\overline{W}_t$$

Diffusion generation refines a joint field.

Classical denoising and score models learn a field and integrate it; DiffWave refines a complete acoustic waveform with bidirectional convolution. The reverse-time SDE above uses decreasing time. HNN's local, modal and geometric refinements compose through their own laws on the whole field. A linear resolvent is a useful special case, not its universal dynamics.

An internal image is also a generated receiver face.

A reconstructed field may perturb another interior Holon before any exterior product is emitted. Gaze, motion and acoustic radiation likewise expose activity through available coupling. The framework concerns these generated and received differences at every grain; deliberate communication and next-token prediction are application restrictions.

Definition for the operator/receiver relationships. Primary comparisons: DDPM, arXiv:2006.11239; DiffWave, arXiv:2009.09761; score SDEs, arXiv:2011.13456. The September 8 generator/tube synthesis and the supplied September 14 papers remain part of this source route.

The construction is networked by its mathematical relations

The formula and library map now recover the manifold, nonlinear recurrence and preimage geometry together.

Operation	Existing source owner	Role in the same object
Holon / tensor current	HOLON.md; HolonTensorLens; field and bilinear operators	Typed sections, contraction, frame, current and material.
Surface and higher incidence	SimplicialComplex; GradedCausalComplex	Stars, links, hinges, common higher cells and $\partial^2=0$.
Torus, phase and holonomy	analytic_field; implicit; lattice_gauge; StructureConnection	Actual carrier equations, cycles, local phases and ordered current transport.
Attention and derivatives	HolonicAdjointNormalization; HolonicArchitectureCharts; normalized native receiver	State-dependent participation, sigmoid/softmax calculus and complete value/material returns.
Recursive arrival geometry	HolonicRecurrentEcology.FirstArrival; bounded basin owner	Full preimages, first-hit characterization and covariance under a changed chart.
Constitutive body and geometry	Active field/junction/coupled body; discrete_curvature at its own scope	Current, internal storage, material variation and changed incidence.
Scale and compression	Tube/clocked span, mode quotient, FractalPacking	Restriction/rebase and future-compatible compression; separated IFS is one specialization.
Illustrated analytic instance	intrinsic_holonic_flow	One twelve-phase map, linked domains, shared volume cells and recomputed arrival sections.

The conceptual substitution is corrected.

The prior document replaced intrinsic recursive geometry with tiling, and replaced the shared manifold with an execution-column picture. It also made a globally contractive affine reference central, removing the route geometry being discussed. Those examples remain historical bounded controls; they no longer define the model or its main illustrations.

Established-bounded · source-inspected for the owner recovery and document correction. The nonlinear numerical reference is a mathematical/physical construction at its stated scope. Native Athena assembly continues through the existing field/body consumer; this presentation does not substitute the reference for that product.

The correction belongs in the workflow.

The hexis convention now permits fractal geometry in a fixed-dimensional state. The model formula, Holon guide, formal framework, architecture map, development method and operating contract direct recovery through current/connection/incidence and recurrence/preimage together. A picture must identify its receiver and preserve the mechanism it is meant to explain.